

1. $78\%, \frac{\pi}{4}, \frac{4}{5}, \frac{64}{75}, \frac{\sqrt{3}}{2}$

EXAM TIP

Convert each value to a decimal first and arrange them accordingly.

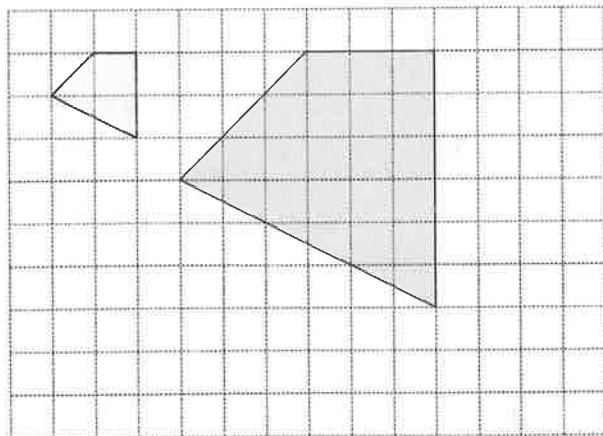
2.

T-shirt	shorts	socks
R	B	S
R	B	P
R	W	S
R	W	P
G	B	S
G	B	P
G	W	S
G	W	P
Y	B	S
Y	B	P
Y	W	S
Y	W	P

EXAM TIP

List all the possible choices systemically.

3.



EXAM TIP

Recall scale factor = $\frac{l_i}{l_o}$, where l_o is the length of a side of the object and l_i is the corresponding length of the image.

4. (a) 16, 10, 4, -2 ...
 $\begin{array}{ccccccc} & \nearrow & \nearrow & \nearrow & & & \\ & -6 & -6 & -6 & & & \end{array}$

5th term: $-2 - 6 = -8$

6th term: $-8 - 6 = -14$

$\therefore$ The next two terms are -8 and -14.

EXAM TIP

Identify the common difference between each consecutive term.

(b) When $n = 3$,

3rd term = $2(3)^2 - 25$

$= -7 \neq 11$

$\therefore$ Heng is not correct.

5. top row: $D y = x + 5$; $A x = 5$

bottom row: $C y = 5$; $F y = -x + 5$

EXAM TIP

An equation of the form $x = a$ is a vertical line. An equation of the form $y = c$ is a horizontal line.

In $y = mx + c$, a positive gradient ($m > 0$) produces a line that slopes upward from left to right; a negative gradient ($m < 0$) produces a line that slopes downward from left to right.

6. $48 \text{ km/h} = \frac{(48 \times 1000) \text{ m}}{(60 \times 60) \text{ s}}$

$= 13 \frac{1}{3} \text{ m/s} > 13 \text{ m/s}$

$\therefore$ Priya travels at a faster speed.

EXAM TIP

Convert the speeds to the same unit.

7. Volume of prism = Area of $\triangle ABC \times BE$

Since $\angle ACB$ is acute, then area of $\triangle ABC < \frac{1}{2}(9.2)(5) = 23 \text{ cm}^2$.

$\therefore$ Volume of prism $< 23 \times 15 = 345 \text{ cm}^3$

The volume of the prism is less than 345 cm^3 .

$\therefore$ Lim is not correct.

EXAM TIP

Recall volume of prism = area of cross section $\times$ height. Here, the height is BE .

8. $\frac{4x}{3} - \frac{5(x+1)}{4} = \frac{16x}{12} - \frac{15(x+1)}{12}$
 $= \frac{16x - 15(x+1)}{12}$
 $= \frac{16x - 15x - 15}{12}$
 $= \frac{x - 15}{12}$

EXAM TIP

Convert the fractions into equivalent fractions with a common denominator (LCM of 3 and 4) to combine them.

9. (a) $a^\circ = 142^\circ$ (vert. opp. $\angle$ s)

$\therefore a = 142$

(b) $b^\circ = 180^\circ - 142^\circ$ (int. $\angle$ s, // lines)

$= 38^\circ$

$\therefore b = 38$

(c) $c^\circ = 180^\circ - 63^\circ - 38^\circ$ (int. $\angle$ s, // lines)

$= 79^\circ$

$\therefore c = 79$

EXAM TIP

Sum of interior angles between parallel lines = 180°

10. (a) $45 : 390 = 3 : 26$

(b) Cost of 10 cakes $= \frac{\$2.52}{6} \times 10$
 $= \$4.20$

11. (a) $720^2 = (2^4 \times 3^2 \times 5)^2$
 $= 2^8 \times 3^4 \times 5^2$

EXAM TIP

Apply $(a^m)^n = a^{m \times n}$.

(b) $HCF = 2^4 \times 3$
 $LCM = 2^7 \times 3^2 \times 5 \times 7^2$
 $\therefore N = 2^7 \times 3 \times 7^2$

EXAM TIP

5 is a factor of 720, but not a factor of N .

12. Radius of semicircle $= \frac{14}{2}$
 $= 7 \text{ cm}$
 Perpendicular height of trapezium $= 7 \text{ cm}$
 $\therefore$ Shaded area $= \frac{1}{2}(9 + 14)(7) + \frac{1}{2}\pi(7)^2$
 $= 157 \text{ cm}^2$ (to 3 s.f.)

EXAM TIP

Area of trapezium $= \frac{1}{2} \times \text{sum of parallel sides} \times \text{height}$

13. (a) $a = \sqrt{3b - c}$
 When $b = 0.08$ and $c = -0.12$,
 $a = \sqrt{3(0.08) - (-0.12)}$
 $= \sqrt{0.36}$
 $= 0.6$

(b) $a = \sqrt{3b - c}$
 $a^2 = 3b - c$
 $3b = a^2 + c$
 $b = \frac{a^2 + c}{3}$

EXAM TIP

To make b the subject of the formula, rearrange the formula such that b is alone on the LHS of the equation.

14. (a) Area of parallelogram $= 9(2.5)$
 $= 22.5 \text{ cm}^2$

EXAM TIP

Area of parallelogram $= \text{Base} \times \text{perpendicular height}$

(b) Area of each square face $= 6^2$
 $= 36 \text{ cm}^2$
 Area of 4 triangular faces $= 85 - 36$
 $= 49 \text{ cm}^2$
 $\therefore$ Surface area of solid B $= 49 + 5(36)$
 $= 229 \text{ cm}^2$

EXAM TIP

Solid B consists of 4 triangular faces and 5 square faces.

15. (a) $\cos 62^\circ = \frac{7}{QR}$

$QR = \frac{7}{\cos 62^\circ}$
 $= 14.9 \text{ m}$ (to 3 s.f.)

EXAM TIP

In a right-angled triangle, $\cos x^\circ = \frac{\text{adjacent side}}{\text{hypotenuse}}$

(b) $\cos 68^\circ = -\cos(180^\circ - 68^\circ)$
 $= -\cos 112^\circ$
 $\therefore x = 112$

EXAM TIP

$\cos x^\circ = -\cos(180^\circ - x^\circ)$

16. Let the radius of the cone be $r \text{ cm}$, i.e. the height is $2.8r \text{ cm}$.
 Volume of cone $= 300 \text{ cm}^3$

$\frac{1}{3}\pi r^2 h = 300$

$\frac{1}{3}\pi r^2(2.8r) = 300$

$\frac{2.8}{3}\pi r^3 = 300$

$r^3 = \frac{900}{2.8\pi}$

$r = \sqrt[3]{\frac{900}{2.8\pi}}$

$= 4.68$ (to 2 d.p.)

$\therefore$ The radius of the cone is 4.68 cm. (shown)

EXAM TIP

Recall volume of cone $= \frac{1}{3}\pi r^2 h$.

17. (a) Number of red cars $= 4 + 3$
 $= 7$

(b) (i) Number of blue cars $= 58 - 20 - 7 - 17$
 $= 14$

(ii) Percentage of cars that are grey $= \frac{20}{58} \times 100\%$
 $= 34.5\%$ (to 3 s.f.)

EXAM TIP

To change $\frac{20}{58}$ to a percentage, multiply $\frac{20}{58}$ by 100%.

18. (a) $\angle BAC = \frac{180^\circ - 144^\circ}{12}$ (base $\angle$ s of isos. $\triangle$)
 $= 18^\circ$

EXAM TIP

$AB = BC$ (sides of a regular polygon)

(b) $\angle BCA = \angle DCE = 18^\circ$
 $\therefore \angle ACE = 144^\circ - 18^\circ - 18^\circ$
 $= 108^\circ$

(c) Join O to E . JOE is a straight line.

$$\begin{aligned}\angle FOE &= \frac{360^\circ}{10} \\ &= 36^\circ\end{aligned}$$

$$\begin{aligned}\therefore \text{Obtuse } \angle FOJ &= 180^\circ - 36^\circ \text{ (adj. } \angle \text{s on a str. line)} \\ &= 144^\circ\end{aligned}$$

$$\begin{aligned}19. (a) \quad AB &= \sqrt{(10-4)^2 + (-2-5)^2} \\ &= \sqrt{85} \\ &= 9.22 \text{ units (to 3 s.f.)}\end{aligned}$$

EXAM TIP

Recall distance between two points $= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$.

$$\begin{aligned}(b) \quad \text{Gradient of } AC &= -\frac{2}{3} \\ \frac{k-5}{-2-4} &= -\frac{2}{3} \\ \frac{k-5}{-6} &= -\frac{2}{3} \\ 3k-15 &= 12 \\ 3k &= 27 \\ k &= 9\end{aligned}$$

EXAM TIP

Recall gradient $= \frac{y_2 - y_1}{x_2 - x_1}$.

20. Mean

$$\begin{aligned}&= \frac{(18 \times 25) + (19 \times 29) + (20 \times p) + (21 \times 19) + (22 \times 15)}{25 + 29 + p + 19 + 15} \\ 19.75 &= \frac{1730 + 20p}{88 + p} \\ 1738 + 19.75p &= 1730 + 20p \\ 0.25p &= 8 \\ p &= 32\end{aligned}$$

EXAM TIP

Apply mean $= \frac{\sum fx}{\sum f}$, where f is the frequency and x is the number of strawberries.

$$21. (a) \quad 5(x-3) = 7x+2$$

$$5x-15 = 7x+2$$

$$-2x = 17$$

$$x = -8\frac{1}{2}$$

$$(b) \quad 3x+4y=1 \quad \text{--- (1)}$$

$$x-2y=-8 \quad \text{--- (2)}$$

$$(2) \times 2: 2x-4y=-16 \quad \text{--- (3)}$$

$$(1) + (3): 5x=-15$$

$$x=-3$$

Substitute $x=-3$ into (1):

$$3(-3)+4y=1$$

$$-9+4y=1$$

$$4y=10$$

$$y=2\frac{1}{2}$$

$$\therefore x=-3, y=2\frac{1}{2}$$

EXAM TIP

Apply the elimination or substitution method.

$$\begin{aligned}22. (a) \quad x &= \frac{65}{100} \times 1.52 \\ &= 0.988 \\ &= 0.99 \text{ (to 2 d.p.)} \\ y &= 1.52 + 0.988 + 1.18 \\ &= 3.688 \\ &= 3.69 \text{ (to 2 d.p.)} \\ \therefore x &= 0.99, y = 3.69\end{aligned}$$

$$(b) \quad \text{Amount excluding GST} = \frac{100}{108} \times \$352.93$$

$$= \$326.7870 \text{ (to 4 d.p.)}$$

$$\text{Amount of electricity used} = \frac{326.7870}{0.2743}$$

$$= 1190 \text{ kWh (to 3 s.f.)}$$

$$\therefore p = 1190$$

EXAM TIP

Find 100%, the amount excluding GST, given that 108% represents \$352.93.

$$23. \quad \frac{3}{x-2} + \frac{8}{x+1} = 5$$

$$\frac{3(x+1)}{(x-2)(x+1)} + \frac{8(x-2)}{(x-2)(x+1)} = 5$$

$$\frac{3x+3+8x-16}{(x-2)(x+1)} = 5$$

$$11x-13 = 5(x-2)(x+1)$$

$$= 5(x^2+x-2x-2)$$

$$= 5x^2-5x-10$$

$$5x^2-16x+3=0$$

$$(5x-1)(x-3)=0$$

$$5x-1=0 \quad \text{or} \quad x-3=0$$

$$5x=1$$

$$x=3$$

$$x = \frac{1}{5}$$

$$\therefore x = \frac{1}{5} \text{ or } x = 3$$

EXAM TIP

Convert the fractions into equivalent fractions with a common denominator to combine them. Then multiply both sides of the equation by $(x-2)(x+1)$.

Paper 2

September/October 2025

Section A

$$1. (a) \quad \text{Modal speed} = 66 \text{ km/h}$$

$$(b) \quad \text{Median} = \frac{59+60}{2}$$

$$= 59.5 \text{ km/h}$$

EXAM TIP

If the number of terms is even, the median is the mean of the two middle values.

$$(c) P(\text{speed less than 50 km/h}) = \frac{13}{50}$$

$$\begin{aligned} 2. (a) \text{ Total amount at the end of 6 years} \\ = \$500 \left(1 + \frac{2.3}{100} \right)^6 \\ = \$573.09 \text{ (to 2 d.p.)} \end{aligned}$$

EXAM TIP

$$\text{Apply total amount} = P \left(1 + \frac{r}{100} \right)^n$$

$$\begin{aligned} (b) \text{ Amount in Kang's account at the end of 6 years} \\ = \$500 \times \frac{2.8}{100} \times 6 + \$500 \\ = \$584 \\ \text{Difference} = \$584 - \$573.0913 \\ = \$10.91 \text{ (to 2 d.p.)} \end{aligned}$$

$$\begin{aligned} 3. (a) \text{ Total number of people} &= \frac{360^\circ}{90^\circ} \times 4200 \\ &= 16\,800 \end{aligned}$$

$$\begin{aligned} (b) \text{ Angle corresponding to the Green Party} \\ = 360^\circ - 90^\circ - 120^\circ \\ = 150^\circ \\ \text{Number of people who voted for the Green Party} \\ = \frac{150^\circ}{360^\circ} \times 16\,800 \\ = 7000 \end{aligned}$$

$$\begin{aligned} 4. (a) \frac{4.936 \times \sqrt{9.7 - 0.96}}{5.765} &\approx \frac{5 \times \sqrt{10 - 1}}{6} \\ &= \frac{5 \times \sqrt{9}}{6} \\ &= \frac{5 \times 3}{6} \\ &= \frac{15}{6} \\ &= 2.5 \end{aligned}$$

$$\begin{aligned} (b) 20.7 \text{ million} &= 20.7 \times 10^6 \\ &= 2.07 \times 10^7 \end{aligned}$$

EXAM TIP

$$1 \text{ million} = 10^6$$

(c) (i) Since $4 < a < 10$ and $4 < b < 10$, then the value $a \times b$ is greater than 10, so $(a \times b) \times 10^n$ is not written in standard form.

$$\begin{aligned} (ii) PQ &= (a \times 10^m) \times (b \times 10^n) \\ &= ab \times 10^{m+n} \end{aligned}$$

$$\text{Since } 16 < ab < 100,$$

$$\text{then } c = \frac{ab}{10}.$$

EXAM TIP

A number is expressed in standard form when written as $A \times 10^n$, where n is an integer and $1 \leq A < 10$, so $1.6 < \frac{ab}{10} < 100$ satisfies this criteria.

$$(iii) PQ = \frac{ab}{10} \times 10^{m+n+1}$$

$$\therefore r = m + n + 1$$

$$\begin{aligned} 5. (a) \text{ Total number of units} &= 4 + 5 + 3 \\ &= 12 \end{aligned}$$

$$\begin{aligned} \text{Maya's share} &= \frac{4}{12} \times \$300 \\ &= \$100 \end{aligned}$$

$$\begin{aligned} \text{Nadia's share} &= \frac{5}{12} \times \$300 \\ &= \$125 \end{aligned}$$

$$\begin{aligned} \text{Rahim's share} &= \frac{3}{12} \times \$300 \\ &= \$75 \end{aligned}$$

EXAM TIP

Based on the ratio, the total amount, \$300, is made up of $4 + 5 + 3 = 12$ units.

$$(b) 8 \text{ cm}^2 \text{ represents } 2 \text{ km}^2.$$

$$1 \text{ cm}^2 \text{ represents } \frac{1}{4} \text{ km}^2.$$

$$1 \text{ cm represents } \frac{1}{2} \text{ km} = 500 \text{ m} = 50\,000 \text{ cm}.$$

$$\therefore n = 50\,000$$

EXAM TIP

Convert the area scale to the linear scale.

$$\begin{aligned} 6. (a) \text{ Distance from } P \text{ to } Q &= x \times 1 \frac{30}{60} \\ &= 1.5x \text{ km} \end{aligned}$$

$$\begin{aligned} \text{Distance from } Q \text{ to } R &= (x - 8) \times 1 \frac{15}{60} \\ &= 1.25(x - 8) \text{ km} \end{aligned}$$

$$\text{Given that total distance} = 111 \text{ km},$$

$$1.5x + 1.25(x - 8) = 111$$

$$1.5x + 1.25x - 10 = 111$$

$$2.75x = 121$$

$$x = 44$$

$$(b) 6x^2 - 7x - 10 = (6x + 5)(x - 2)$$

$$\begin{aligned} (c) ax - 2bx + 3ay - 6by &= x(a - 2b) + 3y(a - 2b) \\ &= (a - 2b)(x + 3y) \end{aligned}$$

EXAM TIP

Factorise by grouping.

$$(d) 3x^2 - 7x - 5 = 0$$

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(3)(-5)}}{2(3)}$$

$$= \frac{7 \pm \sqrt{109}}{6}$$

$$= 2.91 \text{ or } -0.57 \text{ (to 2 d.p.)}$$

EXAM TIP

$$\text{Apply the quadratic formula: } x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

7. (a) Ext. $\angle = 180^\circ - 156^\circ$

$= 24^\circ$

Number of sides $= \frac{360^\circ}{24^\circ}$
 $= 15$

EXAM TIP

Apply exterior angle of regular polygon $= \frac{360^\circ}{n}$, where n is the number of sides.

(b) (i) $BC^2 + BD^2 = 24^2 + 18^2$
 $= 900$

$CD^2 = 30^2$

$= 900$

Since $BC^2 + BD^2 = CD^2$, by the converse of Pythagoras' Theorem, $\triangle BCD$ is a right-angled triangle and $\angle DBC = 90^\circ$. (shown)

EXAM TIP

Determine if Pythagoras' Theorem holds true.

(ii) Area of $\triangle ADC = 337.5 \text{ cm}^2$

$\frac{1}{2}(AB + 24)(18) = 337.5$

$AB + 24 = 37.5$

$AB = 13.5 \text{ cm}$

EXAM TIP

In $\triangle ADC$, AC is the base and BD is the perpendicular height.

(c) The corresponding angles of the two parallelograms are not equal. In parallelograms, having proportional sides alone is not sufficient to guarantee similarity.

EXAM TIP

To conclude that the two parallelograms are similar, all corresponding angles must also be equal. Without this, the shapes may differ in orientation or slant, and thus not be similar.

8. (a) Equation of AB : $x = 2$

EXAM TIP

The x -coordinates of A and B are the same, so AB is a vertical line.

(b) Gradient of $BC = \frac{6 - (-4)}{2 - 6}$
 $= -\frac{5}{2}$

Equation of BC : $y = -\frac{5}{2}x + c$ — (1)

Substitute $x = 2$ and $y = 6$ into (1):

$6 = -\frac{5}{2}(2) + c$

$c = 11$

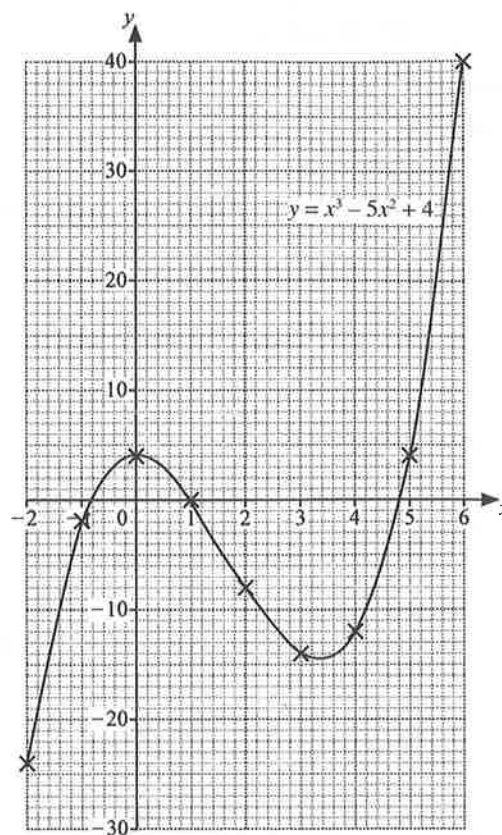
$\therefore$ Equation of BC : $y = -\frac{5}{2}x + 11$

(c) Area of $\triangle ABC = \frac{1}{2}(5)(4)$
 $= 10 \text{ units}^2$

9. (a)

x	-2	-1	0	1	2	3	4	5	6
y	-24	-2	4	0	-8	-14	-12	4	40

(b)



(c) From the graph, when $y = 20$,
 $x = 5.5$.

EXAM TIP

The graph of $y = 20$ is a horizontal line.

10. (a) Let the angle between the ladder and the horizontal be x° .

$\tan x^\circ = \frac{4}{1}$

$x^\circ = 76^\circ$ (to the nearest degree)

$\therefore$ The safe angle between the ladder and the horizontal is 76° . (shown)

(b) (i) Using the ratio 4 : 1,

Distance of the bottom of the ladder from the bottom of the wall

$= \frac{1}{4} \times 5.2$

$= 1.3 \text{ m}$

(ii) Using the ratio 4 : 1,

Distance of the bottom of the ladder from the bottom of the fence

$= \frac{1}{4} \times 1.6$

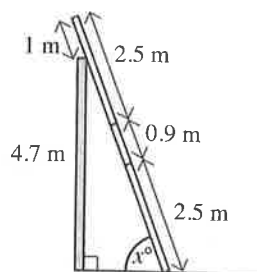
$= 0.4 \text{ m}$

$\therefore$ Distance of the bottom of the fence from the bottom of the wall

$= 1.3 - 0.4$

$= 0.9 \text{ m}$

(c)



- (i) Let the angle between the ladder and the horizontal ground be x° .

$$\sin x^\circ = \frac{4.7}{2.5 + 0.9 + 2.5 - 1}$$

$$= \frac{4.7}{4.9}$$

$$x^\circ = 73.6^\circ \text{ (to 1 d.p.)}$$

$\therefore$ The angle between the ladder and the horizontal ground is 73.6° .

- (ii) 73.6° does not round off to 76° , so the ladder is not placed in a safe position.

Section B

11. (a) (i) Tony may be wrong because he is assuming the extreme values of the intervals are actual data points, and calculates the range as $50 - 10 = 40$ minutes.

EXAM TIP

The minimum possible time is 10 minutes, and the maximum possible time is 50 minutes, so the maximum possible range is $50 - 10 = 40$ minutes.

(ii)

Time, t minutes	Mid-value (x)	Frequency (f)	fx	fx^2
$10 \leq t \leq 20$	15	13	195	2925
$20 < t \leq 25$	22.5	27	607.5	13 668.75
$25 < t \leq 30$	27.5	44	1210	33 275
$30 < t \leq 35$	32.5	38	1235	40 137.5
$35 < t \leq 40$	37.5	19	712.5	26 718.75
$40 < t \leq 50$	45	9	405	18 225
		$\Sigma f = 150$	$\Sigma fx = 4365$	$\Sigma fx^2 = 134 950$

$$\text{Estimated mean} = \frac{4365}{150}$$

$$= 29.1 \text{ min}$$

$$\text{Estimated standard deviation} = \sqrt{\frac{134 950}{150} - 29.1^2}$$

$$= 7.27 \text{ min (to 3 s.f.)}$$

- (iii) School **B** because a smaller standard deviation (6.2 min) means that the times are more consistent, indicating less variation.

EXAM TIP

Apply mean = $\frac{\Sigma fx}{\Sigma f}$, where f is the frequency and x is the mid-value of each interval and standard deviation =

$$\sqrt{\frac{\Sigma fx^2}{\Sigma f} - \left(\frac{\Sigma fx}{\Sigma f}\right)^2}$$

$$(b) (i) P(\text{both balls are white}) = \frac{5}{8} \times \frac{4}{7}$$

$$= \frac{5}{14}$$

$$(ii) P(\text{balls are of different colours})$$

$$= P(\text{black, white}) + P(\text{white, black})$$

$$= \frac{3}{8} \times \frac{5}{7} + \frac{5}{8} \times \frac{3}{7}$$

$$= \frac{15}{28}$$

EXAM TIP

There are two possible cases: black then white, and white then black.

12. (a) Angle $OCB = 18^\circ$

Reasons

$$\angle BOC = 2(72^\circ) \text{ (}\angle \text{ at centre} = 2 \angle \text{ at circumference)}$$

$$= 144^\circ$$

$$\angle OCB = \frac{180^\circ - 144^\circ}{2} \text{ (base } \angle \text{ s of isos. } \triangle)$$

$$= 18^\circ$$

- (b) Using Cosine Rule,

$$QR^2 = PQ^2 + PR^2 - 2(PQ)(PR) \cos \angle QPR$$

$$= 43^2 + 112^2 - 2(43)(112) \cos 110^\circ$$

$$= 17 687 \text{ (to 5 s.f.)}$$

$$QR = \sqrt{17 687} \text{ (} QR > 0 \text{)}$$

$$= 133 \text{ m (to 3 s.f.)}$$

EXAM TIP

Apply Cosine Rule: $c^2 = a^2 + b^2 - 2ab \cos C$.

- (c) Using Sine Rule,

$$\frac{EF}{\sin \angle EDF} = \frac{DF}{\sin \angle DEF}$$

$$\frac{EF}{\sin (180^\circ - 75^\circ - 65^\circ)} = \frac{8}{\sin 75^\circ}$$

$$EF = \frac{8 \sin 40^\circ}{\sin 75^\circ}$$

$$= 5.32 \text{ cm (to 3 s.f.)}$$

EXAM TIP

Apply Sine Rule: $\frac{a}{\sin A} = \frac{b}{\sin B}$.